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$$\Delta H = -3268.12 + \left(6 - \frac{7}{2}\right) \times 8.314 \times 10^{-3} \text{ x } 298$$

$\Delta U =$ -3268.12

$+ (2.5 \times 8.314 \times 10^{-3} \times 298)$

$\Delta U =$ $- 3268.12 + 6.19$

$\Delta U =$ $- 3261.93$ kJ mol^{-1}

4. When a mole of magnesium bromide is prepared from 1 mole of magnesium and 1 mole of liquid bromine, 524 kJ of energy is released.

   The heat of sublimation of metal is 148 kJ mol^{-1}. The heat of dissociation of bromine gas into atoms is 193 kJ mol^{-1}. The heat of vapourisation of liquid bromine is 31 kJ mol^{-1}. The ionisation energy of magnesium is 2187 kJ mol^{-1} and electron affinity of bromine is –331 kJ mol^{-1}. Calculate the lattice energy of magnesium bromide.

Solution :

Given :

$Mg(S) + Br_2(1) \rightarrow$

$MgBr_2(S)$ $\Delta H_f^0 = -524$ KJ mol^{-1}

Sublimation :

$Mg(S) \rightarrow Mg(g)$

$\Delta H_1^0 = +148$ KJ mol^{-1}

Ionisation :

$Mg(g) \rightarrow Mg^{2+}(g) + 2e^-$

$\Delta H_2^0 = 2187$ KJ mol^{-1}

Vapourisation :

$Br_2(1) \rightarrow Br_2(g)$

$\Delta H_3^0 = +31$ KJ mol^{-1}

Dissociation :

$Br_2(g) \rightarrow 2Br(g)$

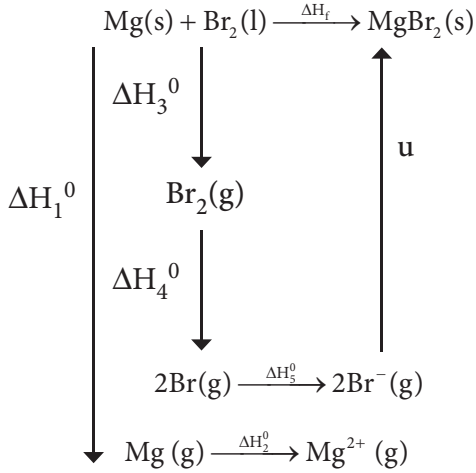
$\Delta H_4^0 = +193$ KJ mol^{-1}

Electron affinity :

$Br(g) + e^- \rightarrow Br^-(g)$

$\Delta H_5^0 = -331$ KJ mol^{-1}

Solution :



$\Delta H_f =$ $\Delta H_1 + \Delta H_2 + \Delta H_3 + \Delta H_4$

$+ 2\Delta H_5 + u$

$-524 =$ $148 + 2187 + 31 + 193$

$+ (2 \times -331) + u$

$-524 =$ $1897 + u$

$u =$ $-524 - 1897$

$u =$ $- 2421$ kJ mol^{-1}

5. An engine operating between 127^0 C and 47^0 C takes some specified amount of heat from a high temperature reservoir. Assuming that there are no frictional losses,